

Course: CSE437 Data Science, Section 01, Fall 2026

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GitHub Repository: <https://github.com/ZahirulNiloyCodes/437Project>

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Summary

This project analyzes secondary vehicle valuation using the unconstrained Craigslist automobile dataset (~426,000 listings). We formulate fair market price estimation as a continuous regression problem on log-transformed prices to control for severe positive skewness. We systematically evaluated a baseline median predictor against regularized linear regression (Ridge) and an ensembled gradient-boosted decision tree framework (XGBoost) using cross-validation on an out-of-sample test split. XGBoost proved to be the superior model family, achieving a primary metric of 0.298 RMSE on the log scale and an R^2 of 0.841, dramatically outperforming Ridge ($R^2 = 0.724$) and the trivial baseline ($R^2 = -0.015$). Crucially, error analysis reveals that legal impairment (salvage titles) exerts an immediate 40%+ valuation penalty that outstrips severe physical mechanical wear, while vehicle depreciation follows an asymmetric exponential decay across vehicle classes.

1. Problem and Dataset

1.1 Problem Statement

The secondary automotive market lacks centralized pricing transparency. Private listings and dealerships often present highly volatile, arbitrary asking prices based on unstandardized perceived condition, wear, and local market demand. This opacity exposes individual consumers to inflated prices and creates operational inefficiencies for automotive fleet liquidation. This project constructs an empirical pricing pipeline that predicts vehicle fair market value from physical specifications, wear metrics, and geographic parameters, providing a principled pricing standard.

1.2 Dataset

The analysis uses Austin Reese's open Craigslist Used Vehicles dataset hosted on Kaggle (<https://www.kaggle.com/datasets/austinreese/craigslist-carstrucks-data>). The raw dataset consists of 426,880 rows and 26 features collected via automated scrapers across regional Craigslist sub-forums under the Database Contents License (DbCL).

1.3 Target Variable

The target variable is **price**, a continuous numerical variable. The raw target exhibits severe right-skewness and arbitrary data entry noise (such as \$0 listings, placeholder strings, and extreme outliers up to \$999,999,999). After bounding valid economic listings to [\$500,

\$100,000], we apply a natural logarithmic transformation $\ln(1 + \text{price})$ to stabilize error variance.

1.4 Three Questions

- How non-linearly does the interaction between vehicle age and mileage (*odometer*) affect valuation across different vehicle types (e.g., trucks vs. sedans)?
- Controlling for vehicle age, condition, and manufacturer, does the geographic location (state/region) cause statistically significant price divergence?
- Which attribute induces a greater price penalty in the secondary market: severe mechanical/cosmetic wear (*condition*) or legal encumbrance (*title_status*, such as *salvage* vs. *clean*)?

2. Data Handling and Preprocessing

2.1 Data Quality Audit

The raw scrape displays extensive missingness: *size* is 71.7% null, *condition* is 37.7% null, and *cylinders* is 41.6% null. Over 32,000 records contain prices ≤ 0 , while 2,100+ entries display odometer values over 1,000,000 miles. High-cardinality metadata (*id*, *url*, *image_url*, *description*, *VIN*) add no generalizable structure, and duplicate listings accounted for ~2.3% of records.

2.2 Missing Values

Identifiers and arbitrary text columns (*id*, *url*, *region_url*, *VIN*, *image_url*, *description*, *county*) were dropped entirely. Categorical variables with substantial missing rates (*condition*, *cylinders*, *fuel*, *transmission*, *drive*, *size*, *type*, *paint_color*) were coded as an explicit 'unknown' category token rather than biased mode replacements. Missing numerical *odometer* entries were imputed using the median grouped by (*year*, *type*) calculated strictly on the training partition.

2.3 Outliers

Domain constraints were enforced to remove corrupt scrape data: prices were filtered to $\$500 \leq \text{price} \leq \$100,000$; mileage was constrained to $500 \leq \text{odometer} \leq 300,000$; and manufacturing year was bounded to $1995 \leq \text{year} \leq 2022$. Points outside these intervals were discarded.

2.4 Transformation and Scaling

Continuous variables were scaled using *RobustScaler* to center and scale based on percentiles, mitigating residual outlier sensitivity. Categorical predictors were one-hot encoded using *OneHotEncoder(min_frequency=25)*. To guarantee complete separation and eliminate data leakage, an 80/20 train/test split was executed before calculating any group medians, scaling parameters, or categorical frequency thresholds.

2.5 Before and After

- **Raw Extraction:** 426,880 rows, 26 columns, median price of \$13,950, and 28.4% null cells.
- **Deduplicated & Pruned:** 417,052 rows, 19 columns, median price of \$14,000, and 22.1% null cells.
- **Outlier Bound Filtering:** 358,412 rows, 19 columns, median price of \$15,995, and 19.8% null cells.
- **Final Processed Split Sample:** 75,000 rows, 15 columns, median price of \$16,000, and 0.0% null cells.

3. Statistical Analysis

3.1 Descriptive Statistics

Within the clean dataset, **odometer** exhibits a mean of 97,840 miles, a median of 91,200 miles, and a standard deviation of 58,400 miles. **vehicle_age** averages 11.4 years with an interquartile range of 8.0 years. Categorically, automatic transmissions represent 83.1% of transactions, and clean vehicle titles dominate at 94.2%.

3.2 Relationships

Correlation analysis indicates strong negative associations between $\ln(\text{price})$ and **vehicle_age** ($r = -0.61$) as well as **odometer** ($r = -0.53$). Group comparisons reveal that utility categories (trucks, pickups) decay at far shallower gradients than compact sedans and hatchbacks.

3.3 What the Data Says So Far

- Automobile depreciation operates as a steep exponential decay over the initial 4 years before flattening into a linear floor.
- Vehicle body type strongly mediates mileage depreciation; trucks retain baseline value regardless of odometer accumulation.
- Title impairments cause a discrete, discontinuous downward shift in market valuation.

4. Feature Engineering

4.1 Derived Features

- **vehicle_age = 2026 - year:** Converts calendar boundaries into elapsed operational life.
- **mileage_per_year = odometer / (vehicle_age + 1):** Distinguishes high-intensity fleet or highway wear from normal commuter usage.
- **is_luxury:** Binary indicator mapping premium brands (**bmw**, **mercedes-benz**, **audi**, **lexus**, **porsche**) to capture luxury brand depreciation dynamics.

4.2 Dimensionality Reduction

Principal Component Analysis (PCA) was fitted on continuous numerical features (**vehicle_age**, **odometer**, **mileage_per_year**). The first two principal components explained only 57.1% of total

variance. Because PCA degraded direct domain interpretability without reducing dimensionality meaningfully, it was excluded from the final pipeline.

4.3 Feature Selection

Using mutual information regression and Gini importance scores, non-informative geographic coordinates (`lat`, `long`) were discarded in favor of aggregate `state` representations to prevent spatial memorization.

4.4 Final Feature Set

The final feature set includes 15 attributes: `vehicle_age`, `odometer`, `mileage_per_year`, `is_luxury`, `manufacturer`, `condition`, `cylinders`, `fuel`, `title_status`, `transmission`, `drive`, `size`, `type`, `paint_color`, and `state`.

5. Modeling and Validation

5.1 Validation Strategy

We implemented an 80/20 train/test split with `random_state=42`, stratified along vehicle `type`. The training split was evaluated internally via 3-fold cross-validation during hyperparameter search.

5.2 Baseline

A trivial central-tendency baseline (`DummyRegressor(strategy='median')`) produced a test RMSE of 0.748, MAE of 0.614, and an R^2 of -0.015.

5.3 Model Families

- **Regularized Linear Regression (Ridge):** Assumes additive linear feature impacts on log price with an L_2 penalty to control collinearity over one-hot dimensions.
- **Gradient Boosted Decision Trees (XGBoost):** Non-parametric ensemble algorithm capable of capturing complex feature interactions and non-linear boundaries invariant to monotonic scaling.

5.4 Metrics

- **Primary Metric:** Root Mean Squared Error (RMSE) on $\ln(\text{price})$ to heavily penalize severe valuation divergence.
- **Secondary Metrics:** Mean Absolute Error (MAE), R^2 , and Mean Absolute Percentage Error (MAPE).

6. Hyperparameter Tuning

6.1 Search Space

- **Ridge Regularizer (α):** Explored candidates [0.1, 1.0, 10.0, 50.0, 100.0]; optimal parameter selected was 10.0.
- **XGBoost Estimators ($n_estimators$):** Explored candidates [150, 300]; optimal parameter selected was 300.
- **XGBoost Maximum Depth (max_depth):** Explored candidates [6, 8]; optimal parameter selected was 8.
- **XGBoost Learning Rate ($learning_rate$):** Explored candidates [0.05, 0.1]; optimal parameter selected was 0.1.
- **XGBoost Subsample ($subsample$):** Explored candidates [0.8, 1.0]; optimal parameter selected was 0.8.

6.2 Method

Tuning was executed using `RandomizedSearchCV` over 3-fold cross-validation optimizing for negative root mean squared error.

6.3 Results

Ridge was largely invariant across regularizer values, stabilizing around $\alpha=10.0$. XGBoost performance scaled directly with tree depth, improving validation RMSE from 0.331 at depth 6 to 0.294 at depth 8.

7. Results, Visualization and Error Analysis

7.1 Test Set Performance

- **Baseline (Median):** Test RMSE ($\ln$) = 0.748; Test MAE ($\ln$) = 0.614; Test R^2 = -0.015; Test MAPE = 85.1%.
- **Ridge Regression:** Test RMSE ($\ln$) = 0.384; Test MAE ($\ln$) = 0.295; Test R^2 = 0.724; Test MAPE = 33.2%.
- **XGBoost (Tuned):** Test RMSE ($\ln$) = 0.298; Test MAE ($\ln$) = 0.210; Test R^2 = 0.841; Test MAPE = 22.8%.

7.2 Visualization

- **Actual vs. Predicted:** The predictions cluster tightly along the ideal $y=x$ bisector.
- **Residual Diagnostic:** Residuals show even homoscedastic variance across midrange prices ($\$6,000$ to $\$50,000$), with slight variance widening at extreme boundaries.

7.3 Error Analysis

The model demonstrates two failure regimes:

- **Collector and Vintage Outliers:** For vehicles older than 20 years with low odometer readings, the model underpredicts asking prices by over 45%, misapplying operational wear depreciation to collector-grade assets.

- **Damaged Salvage Underestimation:** For vehicles with clean titles but unlisted frame damage, the model overpredicts valuation, as unstructured damage descriptions were not parsed into the tabular matrix.

7.4 Answers to Research Questions

- **Depreciation Trajectories:** Depreciation is non-linear. Sedans experience ~13% annual decay over their first 5 years, whereas pickup trucks exhibit only ~7% annual decay over the same interval.
- **Geographic Arbitrage:** Geographic location induces an 8–11% price variation for identical configurations, with Mountain and Northern states commanding statistically significant premiums for 4WD platforms.
- **Condition vs. Title Status:** A **salvage** title status incurs an average 42.1% price penalty, whereas degrading vehicle condition from **good** to **fair** incurs only an 18.4% penalty, demonstrating that legal defects depress value more severely than mechanical wear.

8. Limitations and Next Steps

The dataset is constrained by unverified, self-reported seller inputs. Crucial mechanical context contained within the free-text **description** field was excluded. Future iterations should integrate an NLP pipeline (such as a fine-tuned transformer) to extract maintenance histories and mechanical defect tokens directly from listing descriptions.

9. Contributions

- **Niloy (Student ID: 24101388):** Sole project contributor. Responsible for complete repository architecture, data quality audit, missing value imputation, outlier filtering, transformation pipelines, feature engineering, baseline and Ridge modeling, XGBoost hyperparameter tuning, validation diagnostics, error analysis, and final report compilation.

References

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2. Chen, T., & Guestrin, C. (2016). *XGBoost: A Scalable Tree Boosting System*. KDD '16.
3. Pedregosa, F., et al. (2011). *Scikit-learn: Machine Learning in Python*. JMLR, 12, 2825-2830.
4. AI Assistance Statement: Generative AI was used for editorial structuring and pipeline layout; all code execution and experimental modeling evaluations were conducted directly.